

ELEC4631 Tutorial 1

Do all of the following exercises by hand without using Matlab or any other computational aid.

1. Determine whether the following matrices have some definiteness property (positive definite or semidefinite, negative definite or semidefinite) or not.

(a) $\begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$

(c) $\begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$

2. Find a similarity transformation that diagonalises the following matrix

$$\begin{bmatrix} -3 & 5 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & 4 \end{bmatrix}$$

3. For the matrix A given by

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 4 & 3 & 1 \\ 0 & 0 & 3 \end{bmatrix}$$

determine:

- (a) The characteristic equation and the eigenvalues of A .
- (b) The algebraic multiplicity of each eigenvalue.
- (c) The geometric multiplicity of each eigenvalue.
- (d) Is this matrix diagonalisable? If so, find the matrix that diagonalises it and the diagonalised matrix D .

4. A linear transformation on $\mathbb{R}^2$ is represented on a coordinate system based on the standard basis $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$ by the matrix

$$A = \begin{bmatrix} -1 & 2 \\ 3 & -5 \end{bmatrix}.$$

Suppose that we now switch to a new coordinate system on $\mathbb{R}^2$ that has $f_1 = (2, -1)^T$ and $f_2 = (-1, 1)^T$ as basis. What matrix A' represents the original transformation in this new coordinate system?